Preliminary Proposal Template

Gabrielle Johnson Week 1

Preamble

This week, I was tasked with finding at least five research papers related to AI and machine learning (AI/ML)-driven triage decision-making in emergency departments. There was a plethora of research available on various machine learning techniques, as well as the challenges and considerations associated with implementing these systems in emergency care settings. However, by applying the search strategies learned in this week's tutorial, I was able to significantly narrow down my options and focus on the most relevant studies for my research. I used the following search prompt in the Asta AI tool provided:

"I need to identify peer-reviewed research papers published no earlier than 2020 that investigate the use of artificial intelligence and machine learning algorithms to support patient triage and categorisation in emergency departments. The ideal paper should compare multiple machine learning approaches, evaluate performance across multiple triage systems, utilise multiple datasets or data sources, and use several performance metrics to test validity. The research should be published in a reputable medical or scientific journal and should discuss implementation considerations for low-resource and/or multi-racial healthcare environments, as the findings will be applied to Caribbean emergency departments where electronic health record systems and digital infrastructure remain limited."

By using this targeted search prompt, I was presented with a much narrower and more relevant list of studies. From this refined selection, I was able to identify and select at least three research papers that I believe will provide valuable insights and guidance for the development of my proposed AI-assisted triage system.

Research Paper 1:

Rath, S., Barzani, S., Okesanya, O.J. et al. Optimizing pediatric emergency triage in low-resource settings: evidence-based strategies, task-shifting, and technological innovations. Int J Emerg Med 18, 216 (2025). https://doi.org/10.1186/s12245-025-00998-x

This research paper was selected because it addresses one of the main challenges at Mercer General ED; the impact of limited resources on effective triage. Rath et al.(2025) highlight the negative implications of poor pediatric triage in low-resource settings and provide evidence-based strategies that should be considered when proposing triage improvements. Understanding the existing challenges and possible solutions is important before immediately proposing an AI/ML triage system, because any proposed AI-assisted triage system at Mercer must be practical and appropriate for the low-resource emergency department.

Summary

Rath et al. (2025) identified that pediatric emergency triage in low-resource settings is often weakened by overcrowding, limited staff training, scarce diagnostic tools, poor infrastructure, and triage systems that are not always designed for children. The authors used a comprehensive literature review to examine existing pediatric triage approaches and strategies such as digital decision-support tools, artificial intelligence etc. The study concluded that pediatric triage can be improved through standardized protocols, locally adapted technology etc. However, the paper notes that these solutions may be limited by funding shortages, unequal access to technology, ethical concerns, and the need to adapt tools to each local healthcare setting.

Gaps

One main gap in Rath et al. is that the article focuses primarily on pediatric emergency triage, while Mercer General ED serves both children and adults. Nevertheless, the overall message of the paper is still relevant because many of the challenges discussed, such as overcrowding, delayed treatment, limited staffing and poor infrastructure are not age-specific.

Research Paper 2:

Nadaf H, Jabade MV, Jamadar K, Jogdeo B, Jamdade V. Improving Emergency Response: A Comparative Analysis of Traditional vs Artificial Intelligence-assisted Triage Systems in Health Care and Their Impact. Indian J Crit Care Med 2025;29(11):925–929.

This research paper was chosen because it provides quantifiable evidence comparing AI-assisted triage with traditional triage. Its findings are useful for demonstrating to stakeholders that AI/ML-based triage systems can have measurable benefits, particularly in improving emergency department efficiency. This research supports the argument that resources should be allocated and prioritised to test a similar system in Mercer, where emergency departments also experience delays and resource limitations.

Summary

Nadaf et al. (2025) examined whether AI-assisted triage could improve emergency department time to treatment (TTT) compared with traditional triage systems. The study was conducted in a tertiary hospital in Pune, India, where 105 adult patients were randomly placed into either traditional triage or AI-assisted triage groups. The study found that patients in the AI-assisted group received treatment faster, with an average time of 31.02 minutes compared with 44.12 minutes in the traditional triage group, while ICU admission rates were similar between both groups. However, according to the authors the study was limited by its small sample size, single-hospital setting, short-term focus on TTT and did not examine longer-term outcomes such as death rates, readmission etc. The authors also noted concerns about possible bias, privacy, cybersecurity, transparency, accountability, and the technical and financial difficulty of implementing this type of system in resource-limited hospitals.

Gap

A gap is that the study focused mainly on reducing time to treatment, but did not deeply assess whether the triage decisions themselves were safer or more appropriate. My research can address this by looking at whether basic triage information at Mercer Hospital can help support correct urgency categorisation, not just faster movement through the emergency department.

Research Paper 3:

Leonard F, Gilligan J, Barrett MJ. Development of a low-dimensional model to predict admissions from triage at a pediatric emergency department. JACEP Open. 2022;3:e12779. https://doi.org/10.1002/emp2.12779

This paper was chosen because it demonstrates that a low-dimensional model using routinely collected variables available during the triage process can still accurately predict hospital admission. This is especially relevant to Mercer General ED, where Sister Patrice stated that they do not always have the time or resources to wait for laboratory or radiology results before making triage decisions. Hence, this study supports the idea that basic triage information can be used to assist early decision-making without requiring complex or delayed clinical data.

Summary

Leonard et al.(2022) researched whether pediatric emergency department patients could be admitted or discharged using only data commonly available up to the post-triage stage. The study used 72,229 eligible emergency department attendances from 2017–2018 and followed the CRISP-DM methodology, comparing gradient boosting machine, logistic regression, and naïve Bayes models. The gradient boosting machine performed best, and was then used to identify eight key predictors for a simpler low-dimensional model, which still performed well with an AUC of 0.835. The study showed that admission and discharge probability can be predicted early using limited routinely collected variables, making the approach useful for hospitals with restricted electronic data systems. However, the authors themselves stated that the study was limited by its use of data from a single pediatric emergency department in Ireland and lacked external validation, limited standardisation of some predictors, and stated that further research could look at better understanding false positive and false negative predictions.

Gap

Leonard et al. (2022) developed their model using data from a pediatric emergency department. This creates a gap because Mercer sees both adult and pediatric patients, and the factors that influence triage decisions may differ between age groups. As a result, a model developed only from pediatric data may not be suitable for a hospital that treats a mixed-age population. My research can address this gap by exploring whether a small set of routinely collected triage details can support triage categorisation for both adult and pediatric patients in a low-resource Caribbean emergency department.

Another key gap is that Leonard et al.(2022) relied on admission history, but Mercer may not have digital records that allow this information to be retrieved quickly. My project can address this by exploring and identifying a smaller set of variables that can produce similar or even better results without admission history.

Research Paper 4

Joseph JW, Leventhal EL,Grossestreuer AV, et al. Deep-learning approaches to identify critically Ill patients at emergency department triage using limited information. JACEP Open. 2020;1:773–781. https://doi.org/10.1002/emp2.12218

This paper was chosen because it highlights how routinely collected triage variables can help predict critically ill patients ICU admissions in the emergency department. This is especially important for Mercer General ED, where there are no prior digital records available and where laboratory and radiology results may not be returned quickly enough to support immediate triage decisions. Therefore, the study supports the idea that early, basic patient information collected at triage can still be useful for identifying high-risk patients.

Summary

Joseph et al.(2020) identified that emergency departments need a more accurate way to recognize critically ill patients at triage using only limited information available when patients first arrive. The authors used a retrospective study of adult ED visits and trained machine-learning/deep-learning models using regularly collected triage details such as age, sex, vital signs, ESI score, abnormal vital-sign triggers, and chief complaint text to predict death or ICU admission within 24 hours. The best-performing model combined structured triage data with chief complaint text and achieved an AUC of 0.851, performing better than traditional methods. However, the study was limited because it was retrospective, conducted at a single hospital, and focused mainly on ICU admission,which can be subjective and changes from hospital to hospital.

Gap

The study was conducted at an urban tertiary care hospital, which sees a higher proportion of critically ill patients. Hence, this creates a gap because the model's performance may not transfer directly to Mercer General ED, where the patient population may include a broader mix of low, moderate and high-acuity cases. My project can address this by examining whether limited triage variables are still useful for identifying and classifying all types of patients in a general, low-resource Caribbean ED setting.

Research Paper 5:

Maitham Jawad Aljubran, Hussain Jawad Aljubran, Mohammad Aljubran, Mohammed Alkhalifah, Moayd Alkhalifah, Tawfik Alabdullah Saudi J Er Med. 2023;4(2):112-119
https://doi.org/10.24911/SJEMed/72-1673885230

This paper was chosen because it demonstrates how routinely collected triage data can be used to classify patients into three categories. This is relevant to Mercer General ED because it suggests that basic information already collected during triage may be useful for supporting patient categorisation. The study therefore provides evidence that an AI-assisted triage model could be developed using practical, readily available variables rather than relying on lab/radiology results and past data that can be hard to gather from a paper-based system.

Summary

Aljubran et al. (2023) addressed the issue that traditional emergency department triage depends heavily on clinical judgement, which can lead to incorrect triaging, delays, and inconsistent patient outcomes. To address this problem, the authors used past emergency department records from 2016 to 2019 at a large specialised hospital in Saudi Arabia. Routine triage details were used to classify patients into three urgency groups: nonurgent, urgent, and emergent. The study found that the best-performing model was the ensemble machine learning algorithm CatBoost, which achieved an F1-score of 93.1% and an agreement score of 0.8623. However, the study was based on one specialist hospital with a specialised adult patient population, which may limit how well the findings apply to general emergency departments. In addition, the paper lacked detailed clinical outcome data, so it could not fully show whether the triage classifications led to better patient outcomes.

Gap

One gap in Aljubran et al. (2023) is that the study was conducted in one large adult specialist hospital in Saudi Arabia. This limits how well the findings can be applied to Mercer's smaller, more general emergency department with a wider age range and a more general patient population. My project can address this gap by examining whether routinely collected triage details can still support triaging in Mercer's low-resource emergency department setting.

Research Paper 6:

Chang, YH., Lin, YC., Huang, FW. et al. Using machine learning and natural language processing in triage for prediction of clinical disposition in the emergency department. BMC Emerg Med 24, 237 (2024). https://doi.org/10.1186/s12873-024-01152-1

This paper was chosen because it demonstrates the value of using both structured triage information and unstructured nurse triage notes to improve prediction of patient triaging. This is especially relevant to Mercer General ED, where nurse notes may contain important clinical details that are not captured in fixed triage fields.

Summary

Chang et al. identified that ED triage decisions are often subjective and can lead to incorrect triaging, which affects patient safety and resource allocation. To address this, they developed machine learning models using both structured triage data and NLP-processed free-text nursing notes from two hospitals in Taiwan. The models predicted critical outcomes such as ICU admission, as well as general ward admission/transfer, and performed better than emergency physicians and triaging only prediction. The study found that combining structured data with unstructured clinical notes improved prediction, but limitations included missing data, limited patient history, and reduced generalizability because only two hospitals were included.

Gap

One gap is that the study processed triage notes in a Taiwanese hospital context, including Chinese and English medical terms, which won't be directly transferable to Caribbean dialects and local documentation styles. This is important for Mercer General ED because nurse notes may include informal phrasing, local expressions, abbreviations etc. My project can address this by examining whether nurse-written notes in a Caribbean ED can be standardized or processed in a way similar to the multilingual context above and support triage categorisation.

Literature Gaps

Gaps in the literature relevant to Mercer General ED

Although all the reviewed literature contains gaps that must be addressed before finding an AI/ML solution that can fully fit Mercer General ED, the two gaps listed below are the most realistic and actionable within a 12-week study. Addressing these gaps can provide a practical starting point for developing an AI/ML assisted triage system for Mercer General ED.

Both Aljubran et al. (2023) and Joseph et al. (2020) conducted their studies in large adult specialist hospitals. This limits how well their findings can be applied to Mercer General ED, which is a smaller, lower-resourced ED that serves patients across a wide range of ages and acuity levels. My project can address this gap by examining whether routinely collected triage details can still support accurate triaging in a general Caribbean ED.

Another gap is that Leonard et al. (2022) focused only on pediatric patients and used admission history as part of its model. This may not suit Mercer, which treats a mixed-age population and does not have access to prior digital admission records during triaging. My project can address this by testing whether a smaller set of routine triage variables without patient history, can support triage categorisation in a mixed-age, low-resource general ED.

Problem Statement

Mercer manages approximately 70–95 patients daily, its current paper-based triage causes estimated delays of 85% and 20% category disagreement. Leonard et al.(2022), Joseph et al.(2020), and Aljubran et al.(2023) show that limited ED triage data can support patient-risk prediction and three-level triage categorisation; however, Leonard et al.(2022) focused on pediatric admission, Joseph et al (2020) on patients in a high-acuity ED, and Aljubran et al.(2023) used specialised adult patient data. This 12 week pilot will identify the smallest variable set needed to support triaging at Mercer General ED with ≥90% accuracy and ≤5-minute decision time.

Proposed Solution

Mercer General ED manages approximately 70–95 patients daily, rising to as many as 170 on heavy days, with estimated delays of 85% and approximately 20% triage category disagreement. Evidence from Nadaf et al. (2025) supports testing AI-assisted triage, as their study of 105 adult patients found that AI assisted triage reduced average time to treatment from 44.12 minutes to 31.02 minutes, a reduction of about 13.1 minutes per patient.

Based on these insights, the Mercer pilot will focus on developing a simple AI assisted model using basic triage variables that are already collected during the triage process. The aim is not to replace nurses, but to support their decision making by providing a suggested triage category based on available patient information.

The model will use routinely collected triage variables such as age, sex, chief complaint, mode of arrival, vital signs, pain score, and nurse-assigned triage category. This approach is supported by Leonard et al. (2022), whose simplified model used only eight key predictors and achieved an AUC of 0.835, meaning it had strong ability to distinguish between patients likely to be admitted and those likely to be discharged. Joseph et al. (2020) also showed that limited triage data and chief complaint text could help identify critically ill patients early, achieving an AUC of 0.851 before laboratory or radiology results were available. Aljubran et al. (2023) further showed that routine ED data could classify patients into three urgency groups: nonurgent, urgent, and emergent, with CatBoost algorithm achieving an F1-score of 93.1%, meaning the model had strong overall balance between correctly identifying triage categories and avoiding incorrect classifications. Its agreement score of 0.8623 also suggests strong consistency with existing physician triage decisions.

Leonard et al (2022), Joseph et al. (2020), and Aljubran et al.(2023) provided measurable evidence that limited triage data can support triaging. Hence, the pilot will begin by reviewing Mercer's paper-based triage forms and identifying relevant variables. Afterwards, the data will be digitised, cleaned, and compared with several machine learning models to determine which one is most accurate, fast, and practical for Mercer's setting. The goal is to identify the smallest variable set that can support triaging with at least 90% accuracy and provide a suggested category within five minutes which will reduce delays by 60%, and support nurse decision-making without requiring expensive infrastructure or diagnostic data.

References

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